

Solution

FORCE-01

Class 09 - Science

Section A

1. (a) inertia

Explanation: Inertia is the resistance of any physical object to any change in its state of motion. This includes changes to the object's speed, direction, or state of rest. Inertia will result in falling off the bag from the top of the van.

2. (a) $3v$

Explanation: Due to the conservation of momentum a collision between two bodies, the total momentum of the colliding bodies before the collision is equal to the total momentum after the collision.

$$m_1 v_1 = m_2 v_2$$

$$m_1 = 1, m_2 = 3, v_2 = v$$

$$V_1 = 3v$$

- 3.

(c) $F = \frac{P}{t}$

Explanation: $F = \frac{P}{t}$ because momentum can be given as the product of the force applied and time.

- 4.

(c) two

Explanation: Force is directly proportional to the acceleration.

- 5.

(c) 0.5 m/s^2

Explanation: $F = m \times a$

$$a = F/m$$

$$\text{So, } a = 0.5 \text{ m/s}^2$$

- 6.

(b) N - s

Explanation: The Newton second (also newton-second, symbol Ns or N·s) is the derived SI unit of Momentum. It is dimensionally equivalent to the momentum unit kilogram metre per second (kgm/s). One newton second corresponds to a one-newton force applied for one second.

$$\vec{F} \cdot t = \Delta m \vec{v}$$

- 7.

(d) 4

Explanation: $v = \frac{F \times t}{mass} = \frac{12 \times 1}{3} = 4 \text{ m/s}$

- 8.

(c) 6 m/s^2

Explanation: For the first body,

$$F = m_1 a_1$$

$$\text{So, } 5 = m_1 \times 8$$

$$\text{So, } m_1 = \frac{5}{8} \text{ kg}$$

For second body,

$$F = m_2 a_2$$

$$\text{So, } 5 = m_2 \times 24$$

$$\text{So, } m_2 = \frac{5}{24} \text{ kg}$$

Combined mass of both bodies, $m_1 + m_2$

$$= \frac{5}{8} + \frac{5}{24} = \frac{20}{24} \text{ kg}$$

$$\text{Now, } m = \frac{20}{24} \text{ kg}$$

$$F = 5N$$

$$a = ?$$

$$F = ma$$

$$\text{So, } 5 = \frac{20}{24} \times a$$

$$\text{So, } a = 5 \times \frac{24}{20}$$

$$\text{So, } a = 6\text{m/s}^2$$

9. (a) Both (i) and (ii) are correct.

Explanation: For a lift moving downward with acceleration a , Acceleration of body moving = $g - a$

10.

- (b) The forces acting on the ball are balanced.

Explanation: When all the forces are balanced or the net force is zero, objects move with steady speed.

11.

- (b) Statement 1 is true but statement 2 is false.

Explanation: Force of action reaction never acts on one body. Truck does not move because the push applied is far less than the force of friction between the truck and the road.

12.

- (d) third law of motion

Explanation: The jet engine sucks air in at the front with a fan. A compressor raises the pressure of the air. The compressor is made with many blades attached to a shaft. The blades spin at high speed and compress or squeeze the air. The compressed air is then sprayed with fuel and an electric spark lights the mixture. The burning gases expand and blast out through the nozzle, at the back of the engine. As the jets of gas shoot backward, the engine and the aircraft are thrust forward (Newton's third law of motion).

13. (a) Third law of motion

Explanation: By using Newton's third law of motion, he can get himself to shore. When the person pushes forward, the reaction force offered by the ice is very little. But, this reaction force will help him to move forward. Hence, Newton's Third Law will help him to reach the shore.

14.

- (c) to resists any change in its state of motion

Explanation: Inertia resists any change in its state of motion. It is a property of matter by which it continues in its existing state of rest or uniform motion in a straight line, unless that the state is changed by any external force.

15.

- (d) Newton's third law of motion

Explanation: Inflated balloon lying on the surface of a floor moves forward when pierced with a pin due to Newton's third law of motion. The air coming out of the balloon forces the balloon in the opposite direction.

16.

- (b) always act on different bodies in opposite directions

Explanation: According to Newton's 3rd law of motion, action and reaction are of equal magnitude and directed in opposite direction.

17.

- (b) 5 m/s

Explanation: $IF = \frac{m \times v}{t}$

$$10 = \frac{10 \times v}{5}$$

$$v = \frac{50}{10} = 5\text{m/s}$$

18.

- (b) 8 N

Explanation: In the given problem,

Mass of gun = 20kg ; velocity of bullet = 400m/s

mass of the bullet = 20g = 0.02 kg

$F = \text{change in momentum} / \text{time taken}$

For $t = 1\text{sec}$.

$$F = m \times v / t$$

$$= 400 \times 0.02 / 1$$

$$= 8 \text{ N}$$

19. (a) moves with constant velocity

Explanation: If an object experiences a net-zero unbalanced force, then the body moves with constant velocity. Zero unbalanced forces produce no acceleration in the body and the body continues to move with the same velocity.

20. (a) inertia

Explanation: Inertia is the resistance of any physical object to any change in its state of motion. This includes changes to the object's speed, direction, or state of rest.

- 21.

- (d) move forwards

Explanation: Water moves forward due to inertia of motion. Inertia is an inherent property of an object to resist any change in its state of rest or of uniform motion.

- 22.

- (d) 1.5 N

Explanation: According to Newton's third law, the reading shown by B should be same as A, i.e., 1.5 N.

- 23.

- (d) 0.5

Explanation: Impulse can also be expressed as the rate of change of momentum.

$$\text{And Momentum} = \text{force} \times \text{time} = 5 \times 0.1$$

$$= 0.5 \text{ kg m/s}$$

24. (a) decrease the mass to half

Explanation: Acceleration = force / mass

If force is to be kept constant then, mass must be reduced to half to make acceleration double.

- 25.

- (c) momentum

Explanation: The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object.

- 26.

- (c) 60000 N

Explanation: Here, rate of burning fuel, $\frac{dm}{dt} = 1 \text{ kg s}^{-1}$

$$\text{Velocity of fuel ejection, } v = 60 \text{ km s}^{-1} = 60000 \text{ m s}^{-1}$$

$$\text{Thrust} = \frac{dm}{dt} \cdot v = 1 \times 60000 = 60000 \text{ N}$$

Thus, force exerted on rocket is 60000 N.

27. (a) Both will stop at the same distance

Explanation: As both are moving with same velocity and retardation provided by brakes is equal, so distance travelled would be same for both motorcycle and car.

- 28.

- (d) It always goes away from the earth

Explanation: It always goes away from the earth

- 29.

- (d) A, B

Explanation: Newton's first law deals with rest or uniform motion of a body, i.e.; An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

- 30.

- (c) your school bag

Explanation: Inertia is the resistance of any physical object to any change in its state of motion. This includes changes to the object's speed, direction, or state of rest. Inertia is proportional to mass. Big masses resist changing their motion more than smaller masses. Schoolbag has the largest inertia because the heavier body has larger inertia.

31.

(c) 30 m s^{-2}

Explanation: Let m be the mass of the object.

Ejected mass of the rocket = $\frac{m}{40}$

According to the law of conservation of momentum,

$$mu_1 = \frac{m}{40}u_2; \text{ at } = \frac{u_2}{40}; a = \frac{1200}{40 \times 1} = 30 \text{ m s}^{-2}$$

($\because u_2 = 1200 \text{ m s}^{-1}$ and $t = 1 \text{ second}$)

32.

(d) a cup of tea

Explanation: Inertia is directly proportional to mass and one which poses least resistance also possess least inertia.

33.

(c) accelerated

Explanation: Coin falls behind due to inertia of motion. Inertia is an inherent property of an object to resist any change in its state of rest or of uniform motion.

34.

(c) three

Explanation: If two unequal forces are taken then given two vectors of unequal length, adding them will result in a non-zero vector because, in order to have a zero vector sum, the first and second vectors must "cancel out." For example, if you travel left 1 metre you must travel 1 metre right to return to your initial position. So, 3 unequal forces are mandatory for making resultant force zero.

35.

(b) 1000 N

Explanation: Mass of bullet = $10 \text{ g} = \frac{10}{1000} \text{ kg}$.

initial velocity (u) = 0 and final velocity (v) = 300 m/s.

time (t) = 0.003 s

$$\begin{aligned} F &= \frac{(mv - mu)}{t} \\ &= \frac{(0.01 \times 300 - 0.01 \times 0)}{0.003} \\ &= \frac{(3 - 0)}{0.003} \\ &= \frac{3}{0.003} \\ &= 1000 \text{ N} \end{aligned}$$

36.

(b) momentum

Explanation:

Momentum remains conserved in absence of any external force. Momentum is the product of the mass and velocity of an object. It is a three-dimensional vector quantity, possessing a magnitude and a direction.

37.

(b) on different bodies

Explanation: Reaction from a body is caused due to action of some another body on it.

38.

(c) force

Explanation: Fireman holds a hose by exerting a force because equal and opposite force acts on him. To counter that force he needs to hold the hose with force.

39.

(d) 6.67 ms^{-2}

Explanation: $m_1 = \frac{5N}{10ms^{-2}} = 0.5 \text{ kg}$, $m_2 = \frac{5N}{20ms^{-2}} = 0.25 \text{ kg}$

Total mass = $0.5 + 0.25 = 0.75 \text{ kg}$.

Acceleration produced when both masses are tied together = $\frac{5N}{0.75kg} \text{ kg}$
 $= 6.67 \text{ ms}^{-2}$.

40. (a) Newton

Explanation: The newton (symbol: N) is the International System of Units (SI) derived unit of force. It is named after Isaac Newton in recognition of his work on classical mechanics, specifically Newton's second law of motion.

41.

(b) 30 N

Explanation: Given $m = 200 \text{ g} = 0.2 \text{ kg}$, $u = 60 \text{ m/s}$, $t = 0.4 \text{ s}$

$$\text{Here, } a = \frac{v-u}{t} = \frac{0-60}{0.4} = 150 \text{ m s}^{-2}$$

$$\therefore F = ma = 0.2 \times 150 = 30 \text{ N}$$

42.

(c) All have equal inertia

Explanation: Inertia is directly proportional to mass, hence all the above given objects will have equal inertia.

43.



Explanation: Net external force is zero, it means, particle is moving with constant velocity. Velocity-time graph of such a particle is represented by a straight line parallel to time axis.

44.

(d) 1000 N

Explanation: The force acting on a body is equal to rate of change of momentum. Change of momentum = 10 Kg/s , time taken = 0.01 second . Force = $\frac{10}{0.01} = 1000 \text{ N}$.

45.

(c) its magnitude and direction

Explanation: A force can cause an object with mass to change its velocity (which includes beginning moving from a state of rest), i.e., to accelerate. Force can also be described intuitively as a push or a pull. A force has both magnitude and direction, making it a vector quantity.

46.

(c) 1 : 10

Explanation: $\frac{\text{Acceleration of first block, } a_1}{\text{Acceleration of second block, } a_2} = \frac{1}{10}$

$\therefore$ Force is same on both the blocks.

$$\therefore F = m_1 a_1 = m_2 a_2$$

$$\frac{a_1}{a_2} = \frac{m_2}{m_1} \therefore \frac{m_2}{m_1} = \frac{1}{10}$$

47.

(b) zero

Explanation: No reaction occurs between the ball and the base of the cup.

48.

(c) Force applied by Y on Z is equal to net force on X.

Explanation: Let a be the common acceleration, then $F = 3ma$ or $a = \frac{F}{3m}$

$$\text{Net force on X} = ma = m \times \frac{F}{3m} = \frac{F}{3}$$

$$\text{Force applied by X on Y} = (m + m)a = 2m \times \frac{F}{3m} = \frac{2F}{3}$$

$$\text{Force applied by Y on Z} = ma = m \times \frac{F}{3m} = \frac{F}{3}$$

49.

(c) 0 N

Explanation: According to the first law of motion, an object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced external force

50.

(c) increase the time to slow down

Explanation: To increase the time to slow down so that jerk is minimised and injury is avoided.

51. **(b) N-s**
Explanation: If we multiply the force acting on an object by the time it is acting for this is called the impulse of a force. Impulse is a vector quantity and its unit is the kilogram meter per second (kgms^{-1}) or the newton second (Ns).
52. **(d) decrease the rate of change of momentum**
Explanation: The goalkeeper pulls his hands backwards after holding the ball to decrease the rate of change of momentum by increasing the time. By doing this, less force is exerted on his hands (v Force is directly proportional to the rate of change of momentum).
53. **(a) increase**
Explanation: Speed of truck should be increased to cause minimum loss
- Section B**
54. **(c) Third law of motion**
Explanation: The given image illustrates the third law of motion states that when one object exerts a force on another object, the second object instantaneously exerts a force back on the first. These two forces are always equal in magnitude but opposite in direction. The two opposing forces are also known as action and reaction forces.
55. **(b) A force can change the shape of the object**
Explanation: In the given image, when the softball is pressed in the palm with the help of fingers, the ball changes its shape. This shows that a force can change the shape of the object.
56. **(b) Second law of motion**
Explanation: The image shows that while catching a fast-moving cricket ball, a fielder in the ground gradually pulls his hands backwards. While doing so, the fielder increases the time during which the high velocity of the moving ball decreases to zero. Thus, the acceleration of the ball is decreased and therefore the impact of catching the fast-moving ball is also reduced. Hence, the image correlates with the second law of motion which states that the rate of change of momentum of an object is proportional to the applied unbalanced force in the direction of the force.
57. **(c) A force can stop a moving body**
Explanation: A moving ball comes to a stop when the goalkeeper held his palm in the path of moving the ball. As his palm applies a force to the moving ball. The stopping of a moving football by a goalkeeper demonstrates that a force can stop a moving object.
58. **(b) Frictional force**
Explanation: The force which always opposes the motion of one body over another body is called frictional force or friction. The force of friction always acts on all the moving objects and its direction is always opposite to the direction of motion.
59. **(c) Exerts a forward force on the bullet**
Explanation: When a gun is fired, it exerts a forward force on the bullet. The bullet exerts an equal and opposite reaction force on the gun. This results in the recoil of the gun.
60. **(d) changes its shape**
Explanation: When we press a rubber ball with both the hands, it changes its shape as the force is applied to it.
61. **(a) 12.5 N**
Explanation: $F = \frac{m \times v}{t} = \frac{5 \times 10}{4} = \frac{50}{4} = 12.5 \text{ N}$
62. **(b) Move a stationary object**
Explanation: The given image shows that a force can move a stationary body by applying force on a stationary object.

63. (d) third law of motion
Explanation: As the sailor jumps forward out of a rowing boat, the force on the boat moves it backwards. Thus, when one object exerts a force on another object, the second object instantaneously exerts a force back on the first. Hence, this is an example of the third law of motion.
- Section C**
64. (b) conserve
Explanation: Gun recoil results from the conservation of the total momentum of the bullet-gun system. The backward recoil gun momentum balances the forward bullet momentum to maintain zero total momentum.
65. (a) three
Explanation: Types of Inertia
Inertia of rest: In it a body can't change its state of rest by itself.
Inertia of motion: In it a body can't change its state of motion by itself.
Inertia of direction: In it a body can't change its direction on its own moving in uniform motion.
66. (d) Deform
Explanation: Balanced force is a set of forces acting on a body producing no acceleration in it, but they can deform it.
67. (c) qualitative
Explanation: Qualitative because the first law of motion states that- an object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
68. (b) isolated
Explanation: Force can not generate in a body on its own.
69. (d) motion
Explanation: Momentum (p) = mass $\times$ velocity. or Momentum can be defined as "mass in motion."
70. (c) directly proportional
Explanation: Momentum is directly proportional to the applied force.
Force $\times$ time = momentum
71. (d) motion
Explanation: A high jumper runs for a while before taking high jump so that the inertia of motion helps him in taking a higher jump.
72. (d) inertia
Explanation: Leaves remain at rest and the branch changes its place as a result leaves break off.
73. (c) momentum
Explanation: Momentum changes with change in force, velocity, or acceleration. If no resultant force is there, momentum remains conserved.